

Problem-Solution Fit

Date	01-Jul-2026
Team ID	xxxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

 1. CUSTOMER SEGMENT(S) [CS] - Farmers (Small & Medium Scale) - Agricultural Experts - Researchers - Government Agriculture Departments	 6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited technical knowledge - Budget constraints - Limited internet access in rural areas - Mostly use smartphones for accessing services	 5. AVAILABLE SOLUTIONS PROS & CONS [AS] Available Solutions: - Agricultural experts - Traditional farming knowledge - Government advisory services - Online resources Pros: - Accessible - Experienced guidance Cons: - Time-consuming - Location-dependent - Inconsistent - Not personalized - Lacks AI-based predictions
 2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Difficulty selecting suitable crops (Very High) - Unpredictable weather conditions (High) - Poor soil nutrient knowledge (High) - Reduced crop yield and income (Frequent)	 10. YOUR SOLUTION [SL] OptiCrop OptiCrop is an AI-powered crop recommendation system that analyzes soil nutrients (N, P, K) and environmental parameters (temperature, humidity, pH, rainfall) using machine learning to recommend the most suitable crop. It helps farmers improve productivity, reduce risks, optimize resource utilization, and support sustainable agriculture.	 7. BEHAVIOR + ITS INTENSITY [BE] - Frequently seek advice before planting (High) - Compare different crop options (Medium) - Depend on weather forecasts and local experts (High)
 3. TRIGGERS TO ACT [TR] - Beginning of crop season - Poor harvest in previous season - Soil testing reports - Need to maximize productivity and profit		 8. CHANNELS OF BEHAVIOR [CH] ONLINE - Web application - Mobile browser - Agricultural websites OFFLINE - Agricultural officers - Farmer groups - Local markets - Training programs
 4. EMOTIONS BEFORE / AFTER [EM] Before:  Confused, uncertain, worried about crop failure and financial loss. After:  Confident, satisfied, informed, and optimistic about better yields.		 9. PROBLEM ROOT / CAUSE [RC] - Lack of data-driven decision support - Changing climate conditions - Variations in soil nutrients - Limited access to agricultural expertise

Problem–Solution Fit Canvas

The **Problem–Solution Fit Canvas** is used to understand the challenges faced by farmers and other agricultural stakeholders while selecting suitable crops. It helps identify customer needs, existing problems, limitations, behaviors, and the root causes of poor farming decisions. By analyzing these factors, the OptiCrop system provides an intelligent, machine learning-based crop recommendation solution that supports better agricultural decision-making, improves productivity, and promotes sustainable farming.

1. Customer Segments (CS)

The primary users of **OptiCrop** are **small-scale and medium-scale farmers** who need assistance in selecting the most suitable crop. Other stakeholders include **agricultural extension officers, researchers, agribusiness organizations, and government agricultural departments** that use the system for crop analysis and planning.

2. Problems / Pains and Frequency (PR)

Farmers frequently face difficulties in choosing the right crop because soil nutrients and weather conditions vary from region to region. Incorrect crop selection often leads to **low crop yield, inefficient use of fertilizers and water, reduced productivity, and financial losses**.

3. Triggers to Act (TR)

Farmers seek crop recommendations when preparing for a new cultivation season, after receiving soil test results, following poor harvests, or when weather conditions change significantly. These situations encourage them to look for reliable, data-driven farming advice.

4. Emotions Before / After (EM)

Before using OptiCrop, farmers may feel **confused, uncertain, and worried** about selecting the wrong crop. After receiving AI-based recommendations, they become **confident, satisfied, and optimistic** about improving crop yield and farm profitability.

5. Available Solutions – Pros and Cons (AS)

Existing solutions include advice from agricultural experts, traditional farming experience, government advisory services, and online agricultural resources. While these methods provide useful guidance, they are often **time-consuming, location-dependent, not personalized, and may not consider all soil and environmental parameters**.

6. Customer Limitations (CL)

Many farmers have **limited technical knowledge**, restricted access to agricultural experts, budget constraints, and unreliable internet connectivity. These limitations make it difficult to adopt advanced agricultural technologies and data-driven decision-making.

7. Behavior and Intensity (BE)

Farmers usually consult local experts, discuss with fellow farmers, monitor weather forecasts, and rely on previous farming experience before selecting crops. These activities are performed regularly during every cultivation season.

1. Channels of Behavior (CH)

Online: Crop recommendation website, mobile browser, agricultural portals, weather forecast websites, and government agriculture portals.

Offline: Agricultural extension officers, farmer associations, village meetings, local markets, agricultural training programs, and soil testing centers.

2. Problem Root / Cause (RC)

The main causes of poor crop selection include **variations in soil nutrients, changing climatic conditions, lack of scientific decision support, limited agricultural knowledge, and dependence on traditional farming methods**. These factors reduce productivity and increase farming risks.

3. Your Solution (SL)

OptiCrop is an AI-powered crop recommendation system that uses **Machine Learning** to analyze **Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall**. Based on these parameters, the system recommends the most suitable crop, helping farmers improve productivity, optimize resource utilization, reduce crop failure, and support sustainable agriculture.